Sim-to-Real Transferability and Calibration Requirements for Connected-Vehicle Road Roughness Estimation

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***Abstract*—Connected cars could monitor road roughness continuously, but mapping their vibration to the International Roughness Index (IRI) requires paired drives over profiled roads, and vehicle simulation over existing profiles can generate such pairs. We quantify what transfers from simulation to a production car on public profilometer-referenced data (LiRA-CD, 168 km), testing on held-out routes with spatial-block bootstrap intervals. A randomised half-car simulator driven over measured profiles trains gradient boosting that ranks road segments within routes about as well as real training (Spearman 0.82–0.84 vs 0.83 under nine sensor and vehicle models), yet an untrained, speed-normalised vibration RMS ranks within 0.05 of both on every route: ranking is not the bottleneck. The bottleneck is the absolute level: the zero-shot bias ranges from -0.09 to $+0.43$ m/km and depends on how well the simulated sensor path reproduces the car's vibration above 10 Hz and on the simulated vehicle population. Five kilometres of paired drives resolve it. Simulation with a level offset (pooled MAE 0.42–0.56 m/km) or joint training (0.45–0.47 m/km) beats training on those drives alone (0.68 m/km) for every variant, even with one pass per segment, and the reference simulator with a 5 km offset (0.42 m/km) outperforms 20 km of real drives alone (0.46 m/km). Game-engine data, whose ray-cast road elevation carries 42–54 times the short-wavelength power of measured urban roads, do not transfer.**

***Index Terms*—International Roughness Index, connected vehicles, vehicle vibration, sim-to-real transfer, calibration, road profile.**

I. Introduction

The International Roughness Index (IRI) is the standard summary of longitudinal road roughness [1], [2]. It is computed from a longitudinal profile measured by a laser profilometer, an instrument too costly to survey minor roads often. Production cars that already carry accelerometers could monitor roughness continuously [3], [4], [5], but their vibration depends on the vehicle, the sensor mounting and the speed. The mapping from vibration to IRI must therefore be established for each car type from paired data, drives of that car over road sections whose profile has been measured [6], and collecting such drives is the expensive step.

Vehicle simulation offers another source of pairs. Driving a vehicle model over profiles that road agencies already survey yields vibration with exact IRI labels, at any speed and for many vehicle parameter sets. Estimators trained this way have been validated in simulation [7], [8] and tested with real vehicles for the weighted longitudinal profile [9], for profile reconstruction [10] and for roughness classes [11]; calibrated vehicle models invert real car vibration into road profiles [12], [13] or set an IRI–vibration relation by simulation [14]. What has not been shown, for continuous IRI from a production car on held-out roads of public data, is which parts of a simulation-trained estimator transfer, which simulator choices set its accuracy, and how much paired real data must accompany the simulation.

We answer these questions on LiRA-CD [15], public data of one car model on four routes with profilometer references (Fig. 1). Estimators trained on real drives are compared with the same estimators trained only on simulated drives over the measured profiles of the other routes. We vary the sensor and vehicle model of the simulator, combine it with k km of profiled road with paired drives in three ways, compare simulated and real drives at equal profile budget, in their speed dependence on the same real segments and in their vibration spectra, and benchmark a game-engine corpus against simulation over measured profiles. Our contributions are:

- We show that ranking segments within a route is not the bottleneck: an untrained, speed-normalised vibration RMS ranks them within 0.05 in Spearman correlation of the best trained estimators, real or simulated (Section V-B).
- We show that the absolute level is the bottleneck: it depends on how well the simulated sensor path reproduces the car's vibration above 10 Hz, tested by a band-limited retraining, and on the simulated vehicle population (Sections V-B and V-C).
- We quantify the calibration requirement at equal profile budget and with a level offset, an affine map or joint training: about 5 km of paired drives, with one pass per segment, fix the level, and joint training stays better than real-only training up to 20 km for every simulator variant (Section V-D).
- We measure the speed dependence of real- and

simulation-trained estimators on fixed real roads (Section V-E) and show that simulation must be driven by measured profiles: game-engine road meshes yield an implausible IRI ground truth and do not transfer (Section VI).

Code, configurations and predictions will be released with the published version.

II. Related Work

Response-type measurement and physical vehicle models. IRI is defined on the longitudinal profile through a reference quarter car (the "Golden Car") driven at 80 km/h [1], [16], [2]. Response-type systems measure a vehicle's reaction instead of the profile and must be correlated with a reference profiler for each vehicle [6]. Model-based methods estimate roughness from the responses of a specific vehicle with fixed or identified vehicle models and transfer functions [17], [18]; Bridgelall's road impact factor normalises vertical-acceleration energy for connected vehicles [19]. Wu et al. [14] update a full-car model of the test vehicle from a few impact tests, derive a linear relation between IRI and vibration RMS by simulation over random profiles, and validate it on real roads; this calibrates the vehicle model instead of the estimator. Closest to our data, Skar and Levenberg [12] invert road profiles from the accelerations of ten nominally identical electric cars with a quarter-car model, a PID controller and nonlinear error minimisation, and compare them with a laser profilometer; Agebjär et al. [13] identify a vehicle model of a production car and reconstruct the profile with a Kalman filter. These methods need a model of each vehicle with known or identified parameters; our calibrated quarter-car inversion follows the same idea with a quarter car fitted on the training routes, but it is not a reproduction of [12], whose calibrated vehicle parameters are not available.

Learned estimators on real data. Smartphone studies report agreement with reference roughness that depends on speed, vehicle and phone [20], on mounting [21] and on the number of repeated runs [22]; see also [23], [3], [24]. Data-driven models accounting for speed, vehicle type and mounting were trained on real smartphone data [4], semi-supervised learning exploits unlabelled responses of crowdsourced vehicles [25], [26], and Jeong and Jo [27] validated a deep IRI estimator on data of 29 vehicles against an inertial profiler. On LiRA data, Bajic et al. [28] trained regressors on the car's vertical acceleration and speed along the M3 motorway and tested them on the last 20 % of the same road; we test on routes not used for training and report uncertainty that accounts for spatial dependence.

Learning from simulated responses. Closest in design to our study, Nitsche et al. [9] trained regressors on features of simulated vehicle responses and validated them with a probe vehicle, for the weighted longitudinal profile rather than IRI (according to Bajic et al. [28], over a measured profile, against an inertial profilometer and without varying speed). Networks trained on simulated responses have also been used to reconstruct profiles and classify roughness [29], then, trained on a full-vehicle model over slightly modified measured profiles, applied to that vehicle's measured responses [10], to classify roughness grades from quarter-car simulations [8] and, with a real-vehicle test, from the unsprung-mass response [11], and to estimate IRI without vehicle-specific calibration: Jeong et al. [7] trained CNNs on simulated responses of four vehicle types with real driving-speed profiles and validated them numerically with real road IRI information. Randomising the simulator [30], [31] and adapting the randomisation with a few real roll-outs [32] are the standard devices for simulation-to-real transfer; calibration or joint training with paired drives and the same-segment spectral comparison play this role here. Game engines provide vehicle dynamics for software testing [33], [34], but whether their road meshes give a valid IRI ground truth has not been examined. We add, for continuous IRI from a production car on held-out roads of public data, a decomposition of what a simulation-trained estimator transfers, which simulator choices set its level, and how much paired data must accompany the simulation, for several ways of combining simulated and real data.

III. Data and Simulation

Real cars: LiRA-CD. LiRA-CD [15], [35] contains Renault Zoe cars instrumented with an AutoPi telematics unit (three-axis accelerometer, GPS) connected to the CAN bus and fixed to the frame in the middle console, close to the front axle [15], and surveys of the same lanes by the Danish Road Directorate's P79 profilometer. We use the four routes with car data: two urban roads in Copenhagen (CPH1, part of the O2 ring road, and CPH6) and two motorways (M3, M13), each in both directions. The P79 files provide the wheel-path laser elevations at 0.1 m with GPS; the P79 surveyed CPH1, M3 and M13 in September 2020 and CPH6 in June 2020, and the car passes on these routes were recorded between November 2020 and August 2021, two to fifteen months later [35]. In the car files the accelerometer is logged at a nominal 52.6 Hz (66 % of passes) or 333 Hz, quantised to 0.01 g in most passes, with a median mounting tilt of 10°; speed comes from the CAN bus. After map matching (Section IV) the data contain 16,293 segment-passes over 1,678 distinct 100 m segments (168 km) from 91 passes of at least 8 cars (telematics units), about ten passes per segment. Urban segments have mean IRI 3.36 m/km and are driven at 43 km/h on average, motorway segments 1.16 m/km at 87 km/h. The same cars drive several routes, so leave-one-route-out evaluation is route-disjoint but not vehicle-disjoint for estimators trained on real data (Section V-A tests the consequence).

Labels. Labels are the IRI per 100 m, mean of both wheel paths, recomputed from the P79 profiles with an implementation of the quarter-car algorithm [16], [2] that reproduces the P79's own IRI ($r = 1.000$ over 1,520 segments; an algorithm check, since both come from the same profiles).

Parametric vehicle simulator. A linear four-degree-of-freedom half-car (bounce, pitch, front and rear unsprung masses; validated against a fine-step Runge–Kutta integration,

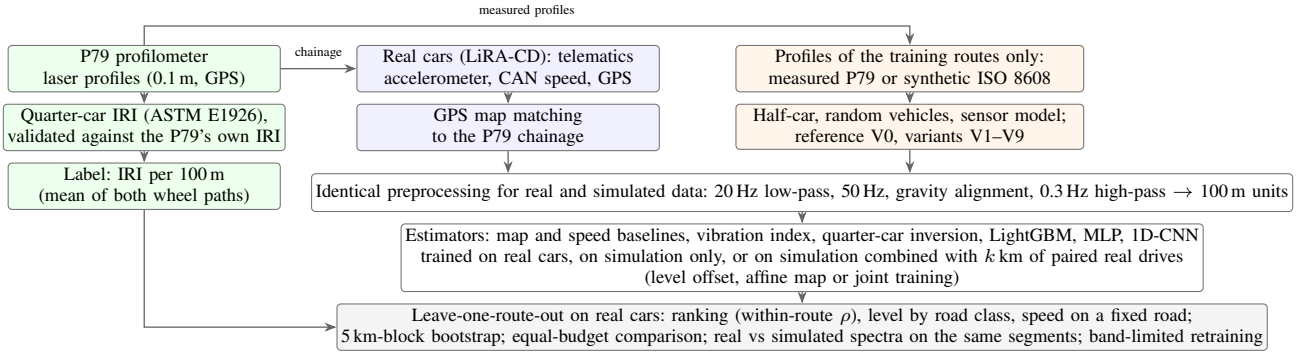


Fig. 1. Study design. Real and simulated units are labelled by the same IRI code, validated against the profilometer's own output, and preprocessed identically; every estimator is evaluated on real routes it never saw, and simulated drives never use the profile of the test route.

Supplement S3) is driven by a road profile under the front axle and the same profile delayed by the wheelbase under the rear axle, after a 250 mm tyre-envelope moving average; on measured profiles the input is the mean of both wheel paths (no roll), and simulated units are labelled exactly like the real ones. Vehicle parameters are drawn from ranges for passenger cars [36] (sprung mass 1000–1800 kg, wheelbase 2.4–3.0 m, body frequency 1.0–1.6 Hz, damping ratio 0.20–0.45; Table S1), speed independently of roughness (20–110 km/h with fluctuations and braking events). The sensor model applies an optional holder resonance, an anti-aliasing filter, a mounting rotation, irregular sampling, scale and bias errors, white noise, quantisation and sample loss.

Simulator variants. The reference simulator **V0** was fixed before any test evaluation: 30 random vehicles, measured P79 profiles of the training routes, and a 50/50 mixture of a telematics-like sensor (55–62 Hz, 0.01 g, rigid mount) and a phone-like sensor (90–210 Hz, 0.6 mg) with a holder resonance at 15–40 Hz. Ten variants change one ingredient each: the telematics-like half alone (**V1**) and the phone-like half alone (**V2**) of V0; the two missing cells of a sensor-family $\times$ holder design, telematics-like with the holder resonance (**V7**) and phone-like without it (**V8**); an ideal sensor (**V3**); a single telematics sensor matched to statistics measured on the unlabelled LiRA-CD streams (**V5**) and its 50/50 mixture with the phone-like sensor (**V6**); one mid-range vehicle instead of 30 (**V4**); and synthetic ISO-8608-type profiles [37] alone (**V9**) or added to the measured ones (**V10**). The simulated drives never use the profile of the test route.

IV. Estimators and Evaluation

Preprocessing and features. Real and simulated accelerometer streams go through the same code: interpolation to 200 Hz, zero-phase 20 Hz low-pass, decimation to 50 Hz, projection on the pass-mean acceleration direction (removing the mounting tilt), and a 0.3 Hz high-pass. LiRA-CD GPS fixes are matched to the P79 track of the same route and direction with a heading constraint. A unit is a 100 m segment traversed once by one pass ($\geq$90 % coverage), described by speed (log mean, coefficient of variation) and vibration features of the dynamic vertical acceleration (log RMS, 95th percentile and maximum of the magnitude, crest factor, kurtosis, band RMS in 0.5–2, 2–5, 5–10 and 10–15 Hz, RMS above 1 Hz, RMS at wavelengths below 5 m).

Estimators. All learned estimators predict log IRI. *Without vibration*: M0, the training median; M0b, a road-class prior that predicts each held-out route with the median IRI of the other route of the same class (needs only a map); M1, LightGBM [38] on the two speed features. *Physics-based*: M2c, a vibration index $\log \mathrm{IRI} = a + b \log(\mathrm{RMS}/v^{\alpha})$ with α fitted; M7, a calibrated quarter-car inversion: the empirical transmissibility of the car and sensor (acceleration spectrum divided by the road displacement spectrum at the pass speed) is estimated on the training routes, a quarter car with sensor gain is fitted to it, each test unit's road spectrum is obtained by spectral division, and IRI follows from the Golden-Car response to that spectrum (this step reproduces the IRI from true profile spectra with $r = 0.94$ in log scale); M7 is also reported with a log-linear output calibration fitted on the training routes. *Learned*: M3, LightGBM on all features (also without speed inputs); M4, an MLP; M5, a 1D-CNN on the resampled acceleration; neural nets are averaged over five seeds. *Untrained*: the statistic $\mathrm{RMS}/v^{0.5}$, a ranking baseline. *Simulation-trained*: M3 and M2c fitted only to simulated units of a simulator variant (zero-shot). LightGBM trained on whole real training routes (M1, M3) selects one of six fixed configurations by inner leave-one-route-out validation; all other LightGBM models (simulation-trained, real-only on k km, joint training) use the first configuration.

Paired data and value of simulation. Paired data are k km of training-route road, drawn as contiguous 1 km blocks (20 random draws per k, identical for every method so that methods are compared draw by draw), with all real passes over them. LiRA-CD drives each 100 m segment about 9.7 times, so $k = 5$ means 475 segment-passes on average (range 301–640), about 47 km of instrumented driving over 4.9 km of profiled road; to test whether this density is needed, at $k = 5$ each segment also keeps at most $m \in \{1, 2, 5\}$ of its passes, drawn at random. These data are combined with an estimator trained on simulated drives over all training-route profiles in

three ways: a level offset (the median log ratio of label to prediction over the paired passes), an affine map in log space, or joint training of LightGBM on the simulated units and the paired passes with equal total weight for the two sets. The real-only comparators are LightGBM and the three-parameter vibration index trained on the same passes. At *equal profile budget*, LightGBM is trained either on the real passes over the k km (which need profiles and paired drives) or on simulated passes over the profiles of the same blocks (which need the profiles only).

Speed on a fixed road. Because each segment is driven about ten times at different speeds, the speed dependence of an estimator can be measured on real roads without confounding by roughness: the within-segment elasticity is the slope of the pass-level log prediction on log speed with segment fixed effects; it should be 0 for a roughness estimator, and $\pm 10\,\%$ between 40 and 100 km/h corresponds to $|\text{elasticity}| \leq 0.104$.

Simulated vs real vibration. Welch spectra of the preprocessed acceleration (2.56 s windows) are compared between real passes and re-simulated passes of four sensor variants and the ideal sensor over the same P79 segments, on cells of segment $\times$ 10 km/h speed band present in both. For all nine measured-profile variants we compare the band RMS features on the same cells (real/simulated amplitude ratio) and the within-segment speed elasticity of the raw RMS (simulated passes restricted to the speed range of the real passes over each segment). To see which inputs carry the differences between variants, the zero-shot LightGBM of every variant is also retrained without the 10–15 Hz band and, separately, on the five features without content above 10 Hz.

Evaluation and inference. The outer split is leave-one-route-out over {CPH1, CPH6, M3, M13}; both directions of the test route are held out together, and no test-route data enter training, calibration or model selection. The evaluation unit is the 100 m segment, predicted by the mean of its pass predictions. We report MAE (primary), bias and Spearman's ρ (also partial ρ given segment speed); *within-route* values are averages over the four routes, *pooled* values are computed over all 1,678 segments, and every MAE is labelled as one or the other. The *level* of an estimator on a route is the mean over its segments of $\log(\text{prediction}/\text{label})$ (0.1 is an over-prediction of about 10 %), and the level of a road class is the mean of the levels of its two routes. Segment errors are spatially correlated (residual autocorrelation 0.44 at 100 m, 0.11 at 1 km), so paired differences between estimators are bootstrapped over 5 km blocks (40 clusters, 2000 resamples), and a difference is called supported only if that interval excludes zero; segment-, 1 km- and route-level intervals are in Table S5; all intervals are conditional on the trained models and use 5 km blocks, except for single-estimator speed elasticities and spectra (1 km). The estimators, the simulator V0, the split and the metrics were fixed before any test result; the variant, calibration, spectral and speed analyses were designed after the test routes had been evaluated and are exploratory (Supplement S1).

V. Results

A. Real cars

Table I lists every estimator on the held-out routes (per route: Table S2). The calibrated quarter-car inversion M7 ranks segments as well as the learned estimators ($\rho = 0.82$) but has a larger error (paired difference to LightGBM $+0.074$ [$+0.039$, $+0.105$] m/km), and its output calibration on the training routes made it worse. LightGBM and the MLP differ negligibly ($+0.004$ [-0.013, $+0.023$]); the 1D-CNN is worse by $+0.045$ [$+0.019$, $+0.075$], almost entirely on CPH1. LightGBM's MAE is 0.60 and 0.70 m/km on the urban routes and 0.13 and 0.15 m/km on the motorways, or 19 % and 12 % of the mean IRI of each class (Table S4). The same telematics units drive several routes, but LightGBM trained only on units that never drove the held-out route is as accurate as on size-matched random subsets (Table S7; the M3 fold is not interpretable), so vehicle overlap did not measurably favour the real-trained estimators (the within-road split of Bajic et al. [28] is in Table S8).

B. Zero-shot transfer: the ranking transfers, the level does not

Trained only on simulated drives over the profiles of the three training routes, the reference simulator V0 reaches a within-route MAE of 0.42 m/km against 0.39 for the real-trained model. The paired difference, $+0.019$ m/km, has a 5 km-block interval of [-0.003, $+0.040$], whereas a segment-level interval, which ignores spatial dependence, would have declared the real-trained model better (Table S5). Both rank segments within routes alike ($\rho = 0.83$ vs 0.83; partial ρ given speed 0.81 vs 0.81). Within-route ranking discriminates little, however, because a Spearman correlation within a route is unaffected by any monotone calibration: the untrained statistic $\text{RMS}/v^{0.5}$ reaches $\rho = 0.82$ (partial 0.81). Paired on the same segments (Table S3), real-trained LightGBM ranks the urban routes better than this statistic (by $+0.042$ and $+0.044$) and the motorways slightly worse (-0.020 and -0.023), and V0-trained LightGBM differs from it by -0.015 to $+0.038$; real-, V0- and V1-trained LightGBM are within 0.05 of it on every route, as are, in point estimates, the other variants, the MLP, the CNN and M7, whereas M2c, LightGBM without speed inputs and the speed-only model rank up to 0.10, 0.18 and 0.72 worse. What training has to supply is mainly the level of the mapping from vibration to IRI.

V0's agreement in level is not a robust property of simulating over measured profiles (Table II, Fig. 2). Across the nine variants on measured profiles (V0–V8), the within-route ρ stays between 0.82 and 0.84 and the scale b between 0.62 and 0.71 (real-trained: 0.66), whereas the pooled zero-shot bias ranges from -0.09 to $+0.43$ m/km. Even V0 matches the level only on average: its pooled bias of $+0.01$ m/km combines a matched urban level (log level $+0.006$) with a motorway over-prediction of about 16 % ($+0.145$; real-trained -0.013 and $+0.016$). In the sensor-family $\times$ holder design, adding the holder resonance changes the bias by -0.35 and -0.36 m/km for the telematics-like and the phone-like sensor,

TABLE I

REAL CARS (LIRA-CD), LEAVE-ONE-ROUTE-OUT, 100 M SEGMENTS. WITHIN-ROUTE METRICS: MEAN OF THE FOUR HELD-OUT ROUTES (SECTION IV); PARTIAL ρ: SPEARMAN CORRELATION WITH IRI GIVEN SEGMENT MEAN SPEED. ΔMAE: PAIRED DIFFERENCE TO THE REAL-TRAINED LIGHTGBM WITH A 95 % CLUSTER-BOOTSTRAP INTERVAL OVER 5 KM BLOCKS (40 CLUSTERS); "ROUTES": NUMBER OF HELD-OUT ROUTES ON WHICH THE ESTIMATOR IS BETTER. NEURAL NETS: 5-SEED ENSEMBLES.

Estimator	Within route			Pooled		ΔMAE vs M3	Routes
	MAE	ρ	partial ρ	MAE	bias	[5 km-block CI]	better
No vibration input							
M0 constant (training median)	1.56	–	–	1.67	−0.64	+1.249 [+1.118, +1.387]	0/4
M0b road-class prior (map only)	0.75	–	–	0.81	−0.20	+0.387 [+0.307, +0.475]	0/4
M1 speed only	0.69	0.39	0.19	0.74	−0.33	+0.324 [+0.252, +0.404]	0/4
Physics-based, calibrated on real training routes							
M2c vibration index	0.62	0.76	0.77	0.67	−0.13	+0.251 [+0.202, +0.306]	0/4
M7 quarter-car inversion	0.46	0.82	0.81	0.49	−0.08	+0.074 [+0.039, +0.105]	0/4
M7 + output calibration	0.54	0.82	0.81	0.58	+0.05	+0.155 [+0.109, +0.201]	0/4
Learned on real cars (3 other routes)							
M3 LightGBM (reference)	0.39	0.83	0.81	0.42	−0.11	ref.	–
M3 LightGBM, no speed input	0.62	0.76	0.72	0.65	−0.24	+0.235 [+0.177, +0.294]	0/4
M4 MLP	0.40	0.83	0.81	0.42	−0.09	+0.004 [−0.013, +0.023]	0/4
M5 1D-CNN	0.43	0.81	0.79	0.47	−0.21	+0.045 [+0.019, +0.075]	1/4
Learned on simulation only (zero-shot)							
V0 LightGBM, measured profiles	0.42	0.83	0.81	0.44	+0.01	+0.019 [−0.003, +0.040]	1/4
V10 LightGBM, measured + synthetic	0.44	0.84	0.82	0.46	−0.04	+0.039 [+0.007, +0.068]	2/4

TABLE II

SIMULATOR VARIANTS: LIGHTGBM TRAINED ONLY ON SIMULATED DRIVES (ZERO-SHOT), TESTED ON THE HELD-OUT REAL ROUTES. POOLED SEGMENT MAE AND BIAS IN M/KM; LEVEL: MEAN LOG(PREDICTION/LABEL) OVER THE SEGMENTS OF A ROUTE, AVERAGED OVER THE TWO ROUTES OF A ROAD CLASS; WITHIN-ROUTE SPEARMAN ρ. RIGHT: POOLED MAE WITH 5 OR 20 KM OF PAIRED DRIVES USED FOR A LEVEL OFFSET (2.5–97.5 % RANGE OVER THE 20 DRAWS IN BRACKETS) OR FOR JOINT TRAINING; MEANS OVER 20 DRAWS. REAL-ONLY LIGHTGBM ON THE SAME DRIVES: 0.68 M/KM (5 KM), 0.46 M/KM (20 KM). SCALE b AND SPEED ELASTICITIES: TABLE S12.

	Zero-shot					+ level offset		+ joint training	
Simulator variant	MAE	bias	level urban	level motorway	ρ	5 km	20 km	5 km	20 km
V0 mixture (reference)	0.44	+0.01	+0.01	+0.14	0.83	0.42 [0.40, 0.45]	0.42	0.45	0.44
V1 telematics-like, rigid	0.68	+0.43	+0.14	+0.40	0.83	0.53 [0.47, 0.60]	0.54	0.47	0.44
V7 telematics-like + holder	0.46	+0.07	+0.02	+0.18	0.83	0.43 [0.40, 0.46]	0.43	0.46	0.44
V8 phone-like, rigid	0.55	+0.27	+0.10	+0.26	0.84	0.45 [0.42, 0.49]	0.46	0.45	0.44
V2 phone-like + holder	0.41	−0.09	−0.03	+0.06	0.83	0.42 [0.40, 0.45]	0.41	0.45	0.43
V3 ideal sensor	0.67	+0.40	+0.13	+0.40	0.83	0.54 [0.48, 0.62]	0.55	0.47	0.45
V5 telematics matched to LiRA-CD	0.69	+0.40	+0.12	+0.42	0.82	0.56 [0.51, 0.64]	0.58	0.47	0.44
V6 matched telematics + phone-like	0.46	−0.04	−0.03	+0.19	0.83	0.45 [0.41, 0.48]	0.45	0.45	0.43
V4 one mid-range vehicle	0.63	+0.40	+0.16	+0.29	0.83	0.44 [0.41, 0.46]	0.43	0.47	0.44
V10 measured + synthetic profiles	0.46	−0.04	−0.04	+0.20	0.84	0.46 [0.41, 0.50]	0.46	0.45	0.43
V9 synthetic profiles only	0.71	+0.36	+0.07	+0.49	0.81	0.75 [0.65, 0.88]	0.77	–	–
trained on real cars (M3)	0.42	−0.11	−0.01	+0.02	0.83	–	–	–	–

and switching the sensor family by −0.16 and −0.17 m/km without and with holder. The vehicle population moves it as much: with V0's sensor mixture, holder resonance included, but one mid-range vehicle instead of 30 (V4), the bias rises by +0.39 m/km. V0's level thus comes from both its holder resonance (meant for phone holders, not the car's telematics unit) and its 30-vehicle population. An ideal sensor (V3) and a telematics sensor matched to the rates, quantisation, noise and tilt of the LiRA-CD streams (V5) both overestimate by about 0.4 m/km. Synthetic profiles alone (V9) still rank segments ($\rho = 0.81$) but overestimate on motorways (bias +0.70 m/km), and adding them to the measured profiles (V10) changes little.

C. Where the simulated vibration differs

On the same segments and speed bands, the real car's vibration differs from all simulated variants in the same way below 10 Hz and differently above (Fig. 3a, Table S19): all variants lack energy at 2–5 Hz (real/simulated power 2.4–3.6) and match it at 5–10 Hz, whereas at 10–15 Hz the real car has 5.2 and 4.9 times the power of the ideal and the rigid telematics-like sensor but only 1.9 times that of the phone-like sensor with holder resonance. Across the nine variants the zero-shot bias orders most closely like the amplitude deficit at 10–15 Hz (Spearman 0.97), but also like those at 5–10 Hz (0.80) and of the overall RMS (0.88), and hardly like those at 0.5–2 and 2–5 Hz (0.22 and 0.25; Fig. 3b, Table S21); a linear fit explains about as much of the bias with the overall RMS ratio as with the 10–15 Hz ratio (R^2 0.78 and 0.71). These associations were found after the fact on variants that are not independent, and V4 departs from them: its 10–15 Hz ratio is close to V0's (1.93 vs 1.86), its bias is not. The deficit is larger on motorway than on urban segments for 9 of 9 variants (Table S20), as is the level error.

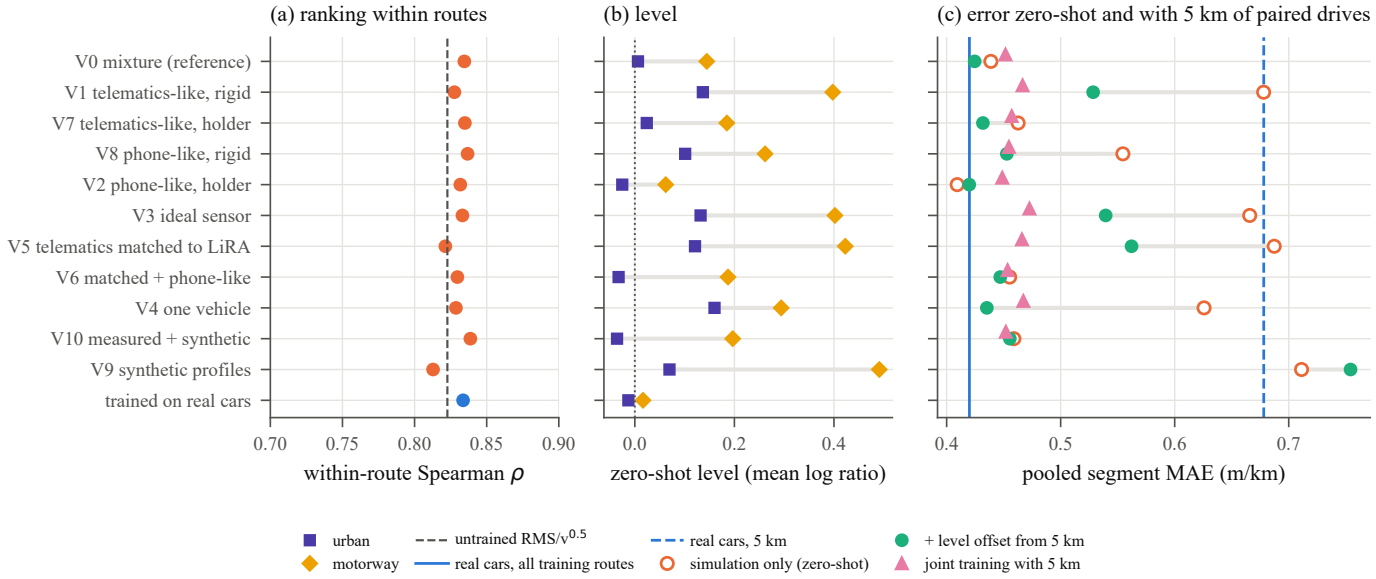


Fig. 2. Simulator variants, LightGBM trained on simulated drives only and tested on the held-out real routes. (a) Within-route Spearman ρ. (b) Zero-shot level (mean log ratio of prediction to label; mean of a class's two routes). (c) Pooled segment MAE zero-shot and with 5 km of paired drives (means over 20 draws).

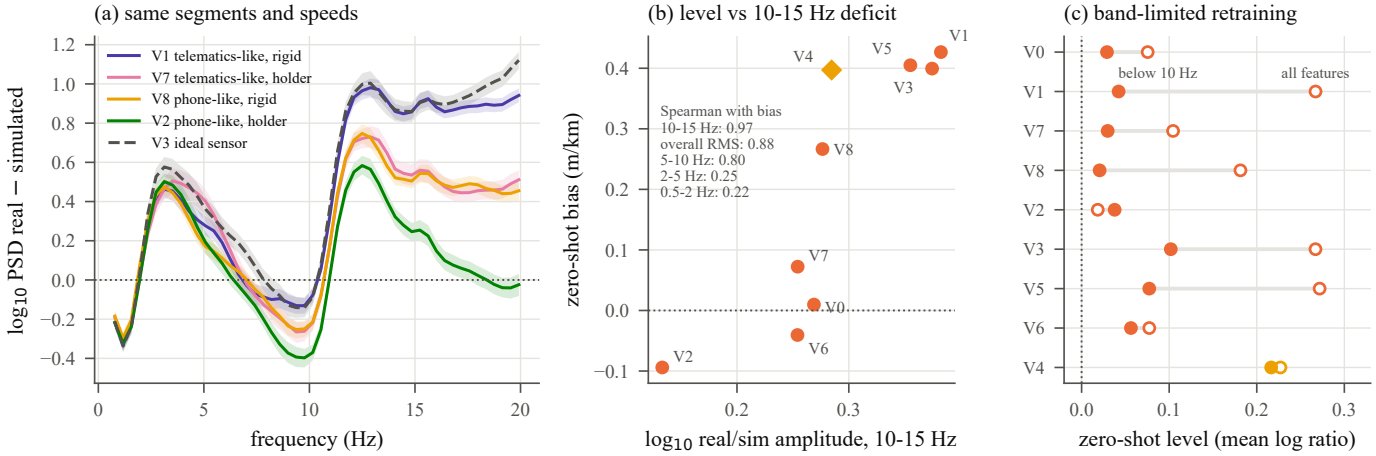


Fig. 3. (a) Mean difference of $\log_{10}$ power spectral density, real minus simulated, on the same segments in the same 10 km/h speed bands (re-simulated passes of four sensor variants and the ideal sensor; bands: 95 % intervals from a 1 km-block bootstrap). (b) Zero-shot bias of the nine measured-profile variants against the real/simulated amplitude ratio of the 10–15 Hz band ($\log_{10}$; V4: one vehicle), with the Spearman correlation of the bias with the ratio of every band. (c) Mean zero-shot level of each variant with all 13 features and with the five features without content above 10 Hz.

To test which inputs carry the differences between the sensor variants, we retrained the zero-shot LightGBM of every variant on the 5 of 13 features that contain no vibration above 10 Hz (speed, speed variation and the 0.5–2, 2–5 and 5–10 Hz bands). Over the eight sensor variants (all but V4), the spread of the mean log level fell from 0.25 to 0.08 (32 %; urban level 49 %, motorway level 34 %) and that of the pooled bias from 0.52 to 0.14 m/km, whereas dropping only the 10–15 Hz band changed every spread by at most 2 % (Fig. 3c, Table S22): the variants differ through all features that carry vibration above 10 Hz, not through the 10–15 Hz band alone. The low-band estimator is not more accurate: for every sensor variant it under-predicts urban and over-predicts motorway segments (log level −0.12 to −0.04 and +0.12 to +0.24), and after a 5 km level offset the pooled MAE of V0–V8 is 0.48–0.57 m/km instead of 0.42–0.56. V4 stays outside the range of the sensor variants (mean log level +0.22 against +0.02 to +0.10), so the vehicle population sets part of the level independently of the vibration above 10 Hz. The test intervenes on the estimator's inputs, not on the car: it shows where the variant differences enter the estimator, not which physical mechanism produces the real car's extra vibration.

The real car's empirical transmissibility peaks at 12.6 Hz (Fig. S2b), and passes logged at 333 Hz contain more, not less, 10–15 Hz power than passes logged at 52.6 Hz on the same cells (52.6 minus 333 Hz: −0.24 in $\log_{10}$, confounded

with the device), so the excess is not an aliasing artefact. Its physical source is not identified: the telematics unit is fixed to the frame in the middle console close to the front axle [15], and the linear half-car omits local modes of the console and its mount, nonlinearity and friction in tyres and suspension, vibration of the electric drivetrain, and roll and body flexibility. A second mismatch is common to all variants: on a fixed road the real car's vibration RMS rises with speed with elasticity 0.60 [0.50, 0.69] on urban and 0.85 [0.74, 0.93] on motorway segments, the simulated vibration over the same segments at the same speeds with 0.92–1.30 and 1.07–1.40 (Table S20); as this mismatch is similar for all variants, it does not explain their different levels.

D. How much paired real data is needed

Fig. 4 and Table III compare simulation with paired real drives on the same 20 draws of k km of training-route road (every variant: Tables S14 and S15). Of the two real-only comparators, the vibration index suits very small samples better (pooled MAE 1.30 vs 1.63 m/km for LightGBM at 1 km). Ranges over draws reflect which k km are drawn; intervals over the test routes are in Table S16.

Equal profile budget (Fig. 4a). Simulated drives over the profiles of the k km are no general substitute for the real drives over them: with 1–2 km, simulated LightGBM beats real-trained LightGBM for every variant but not the real-trained index (better in 3–10 of 20 draws at 1 km), and from 5 km on the answer depends on the variant (at 20 km, -0.02 to $+0.24$ m/km against real-only LightGBM, largest for V1, V3 and V5, the variants with the largest 10–15 Hz deficit); a level offset from the same k km closes the gap of V4 ($+0.01$ m/km) but not theirs.

All profiles plus paired drives (Fig. 4b). Trained on simulated drives over all profiled training roads and combined with 5 km of paired drives, every variant beats training on those drives alone (pooled MAE 0.68 m/km), whether the paired drives set a level offset (0.42–0.56 m/km, better in 13–20 of 20 draws) or enter joint training with the simulated units (0.45–0.47 m/km, better in 20 of 20 draws). The test-route intervals exclude zero for every variant and both combinations (differences -0.26 to -0.12 and -0.23 to -0.21 m/km), within-route MAEs agree (Table III), and the reference simulator with a 5 km offset (0.42 m/km) is more accurate than 20 km (0.46) or 40 km (0.44 m/km) of real drives alone. With more paired data the two combinations diverge. Real-only LightGBM overtakes the level offset of V1, V3 and V5, which wins only 3–5 of 20 draws at 10 km and 0 at 20 km (test-route intervals above zero; differences $+0.08$ to $+0.12$ m/km), whereas joint training stays better than real-only LightGBM for every variant up to 20 km (-0.03 to -0.01 m/km at 20 km, better in 16–20 of 20 draws, every test-route interval below zero). Per held-out route it is better in 31 of the 36 combinations of variant and route at 20 km; the other 5 (4 on M3, 1 on CPH6) differ by at most $+0.016$ m/km. With joint training the variant matters little at 5 and 20 km (the nine variants span 0.02 and 0.02 m/km, against 0.14 and 0.17 m/km with a level offset), but it still matters at 1 km (0.44–0.56 m/km).

Which combination is better depends on the kind of level error. A level offset removes a common level and leaves the difference between the motorway and urban levels, which ranges from 0.09 to 0.30 over V0–V8 (Table II) and orders their MAE after the offset (Spearman 0.88). At 5 km, joint training minus level offset is -0.096 to -0.062 m/km for V1, V3 and V5, whose class difference is largest (0.26–0.30), and $+0.002$ to $+0.032$ m/km for the other six (0.09–0.22). An affine map fitted to the paired passes was less accurate than the better of the two for 9 of 9 variants at 5 km and 9 of 9 at 20 km (Table S15).

LiRA-CD drives each segment about ten times, but with simulation far fewer passes suffice: keeping at most one pass per segment of the 5 km (49 instead of 475 segment-passes on average) hardly changed the combinations with simulation (level offset 0.43–0.55 m/km, joint training 0.44–0.49), which stayed better than real-only LightGBM on the same passes in every draw for every variant, whereas real-only LightGBM degraded from 0.68 to 0.93 m/km (Table S17). The simulation also contributes more than a speed exponent and a slope: a vibration index $\log \mathrm{IRI} = c + b \log(\mathrm{RMS}/v^{\alpha})$ with the car's own speed exponent ($\alpha = 0.59$–0.72, estimated without labels within segments of the training routes) and an affine fit to the same paired passes reaches 0.78 m/km at 5 km and 0.76 m/km at 20 km.

E. Speed dependence on a fixed road

On a fixed road (median within-segment SD of log speed 0.12; Fig. S4, Table S23; 1 km-block intervals for single elasticities), the real car's vibration RMS rises by $+83$ % between 40 and 100 km/h (elasticity $+0.66$ [$+0.60$, $+0.71$]). The estimators trained on real cars over-correct for it: LightGBM's estimate changes by -13 % over the same range (-0.15 [-0.19, -0.11]), also without speed inputs (-0.14), the MLP and CNN behave alike (-0.19 and -0.19), and the vibration index over-corrects most (-0.64). Simulation-trained LightGBM V0 changes by -6 % (-0.06 [-0.11, -0.01]); paired on the same passes, its elasticity exceeds that of the real-trained model by $+0.09$ [$+0.06$, $+0.12$] (5 km blocks), entirely on urban roads ($+0.12$ [$+0.09$, $+0.15$]; motorways $+0.02$ [-0.06, $+0.06$]), and across the nine variants it ranges from $+0.00$ to $+0.21$ (interval excluding zero for 8 of 9). These elasticities change little with telematics-unit fixed effects or with the within-pass speed SD as a covariate (Table S24), except that slow urban passes are more dynamic (within-segment correlation of log speed and log speed SD -0.50): with the speed SD as a covariate the real-trained urban elasticity shrinks to -0.06 [-0.16, $+0.02$], while the paired urban difference stays at $+0.12$ [$+0.09$, $+0.16$]. The speed coefficient of variation, tied to $1/v$ by construction (correlation -0.86 with log speed), absorbs most of the speed variation within urban segments and turns that difference negative (-0.07 [-0.21, $+0.01$]), so on urban roads speed and speed variability cannot be fully separated. The smaller urban over-correction of simulation-

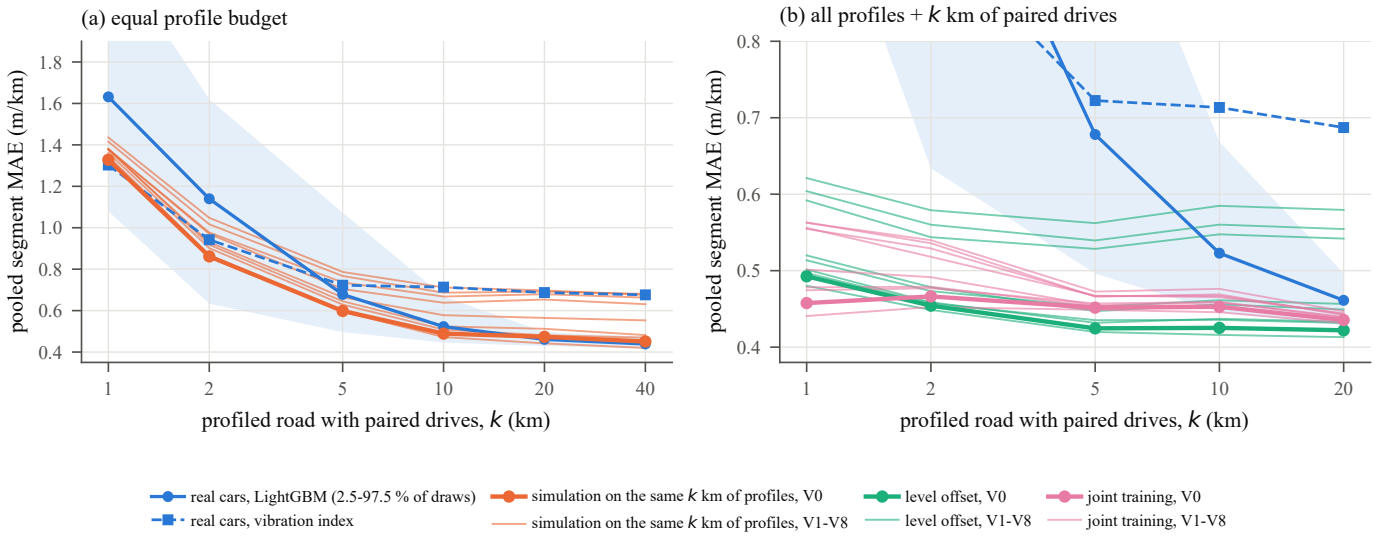


Fig. 4. Simulation against paired real drives on the same 20 draws of k km of training-route road (pooled segment MAE on the held-out routes, means over draws). (a) Equal profile budget: LightGBM on simulated passes over the profiles of the same k km. (b) Simulation over all training-route profiles combined with the paired passes; real-only values at 1 and 2 km lie above the axis (LightGBM 1.63 and 1.14, index 1.30 and 0.94 m/km).

TABLE III

SIMULATION AND PAIRED REAL DRIVES ON THE SAME 20 DRAWS OF k KM OF TRAINING-ROUTE ROAD (PASSES: MEAN NUMBER OF SEGMENT-PASSES). POOLED MAE (BRACKETS: 2.5–97.5 % OF DRAWS) AND WITHIN-ROUTE MAE, IN M/KM. REAL-ONLY: LIGHTGBM OR THE THREE-PARAMETER VIBRATION INDEX ON THE PAIRED PASSES. EQUAL PROFILE BUDGET: LIGHTGBM ON V0 SIMULATED PASSES OVER THE PROFILES OF THE SAME k KM (DIFFERENCE TO REAL-ONLY LIGHTGBM; DRAWS IN WHICH SIMULATION WAS BETTER). ALL PROFILES: SIMULATION OVER ALL TRAINING-ROUTE PROFILES COMBINED WITH THE PAIRED PASSES; RANGES OVER V0–V8 OF THE MEANS, AND OF THE NUMBER OF DRAWS IN WHICH JOINT TRAINING BEAT REAL-ONLY LIGHTGBM.

		Real-only			Equal profile budget (V0)		All profiles + level offset		All profiles + joint training		
k	passes	LightGBM pooled	within	index	pooled	− real (better)	pooled	within	pooled	within	better
1	92	1.63 [1.08, 2.36]	1.572	1.30	1.33	−0.30 (20)	0.48–0.62	0.450–0.576	0.44–0.56	0.415–0.543	20
2	184	1.14 [0.63, 1.62]	1.105	0.94	0.86	−0.28 (20)	0.45–0.58	0.421–0.539	0.45–0.54	0.426–0.519	20
5	475	0.68 [0.50, 1.07]	0.629	0.72	0.60	−0.08 (16)	0.42–0.56	0.395–0.519	0.45–0.47	0.422–0.445	20
10	947	0.52 [0.45, 0.67]	0.487	0.71	0.49	−0.03 (15)	0.42–0.58	0.391–0.537	0.45–0.48	0.418–0.445	18–20
20	1,905	0.46 [0.43, 0.50]	0.433	0.69	0.47	+0.01 (9)	0.41–0.58	0.388–0.532	0.43–0.45	0.405–0.419	16–20
40	3,777	0.44 [0.42, 0.47]	0.412	0.68	0.45	+0.01 (3)	–	–	–	–	–
all		0.42	0.395	–	0.44 (zero-shot)	–	–	–	–	–	–

trained LightGBM is therefore not established, and given the speed-scaling mismatch of Section V-C any difference reflects the learned estimators, not a reproduced speed response of the car.

VI. THE SIMULATOR FIDELITY GAP: GAME ENGINES VERSUS MEASURED PROFILES

The source of the road profiles decides whether simulated training transfers. Driven by measured profiles, the parametric half-car gives the results of Section V; driven by synthetic ISO 8608 profiles (V9), it still ranks segments ($\rho = 0.81$) but over-predicts motorway roughness by $+0.70$ m/km. To measure the fidelity gap of game-engine environments, we benchmark a BeamNG.tech [34] corpus: 19 manually driven trips (138 km) of 3 vehicles on 3 maps, logging the vehicle's inertial signals and the ray-cast elevation of the road mesh under both wheel paths, from which IRI labels are computed with the same quarter-car algorithm after the smoothing discussed below. LightGBM trained on it does not transfer to LiRA-CD: its within-route ρ is 0.35 and its MAE 1.26 m/km (0.32 and 2.08 m/km with unsmoothed labels), and its estimates rise steeply with speed on a fixed road (elasticity $+0.79$; further results in Supplement S2).

The ground truth explains much of this. The ray-cast mesh elevation is far rougher than the measured roads, increasingly so at short wavelengths: its displacement power exceeds that of the urban P79 profiles 12-fold at wavelengths of 10–30 m and 42–54-fold at 0.5–5 m (Fig. 5), and its IRI averages 12.5 m/km, against 3.36 m/km on the urban P79 roads. The excess lies in the elevation values, not in their sampling: 54–69 % of the logged readings repeat the previous one while the car moves, but removing the repeats raises the mean IRI to 15.6 m/km. A curvature-adaptive smoothing designed to remove mesh tessellation lowers it to 4.5 m/km and changes the condition class of 34 % of the segments, yet applied to the measured profiles it lowers the urban IRI by 28 % (Fig. S1b), so it cannot

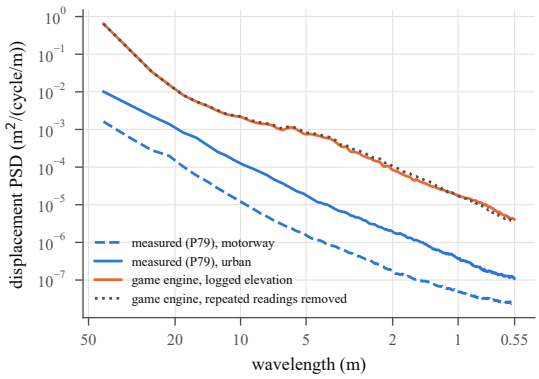


Fig. 5. Displacement power spectral density (Welch, detrended, 0.25 m grid) of the measured P79 profiles (laser 5 and 21) and of the ray-cast mesh elevation of the game-engine corpus (both wheel paths of all trips), as logged and with repeated readings removed.

repair the labels without distorting real roughness. The logged inertial signals are also degraded: 48–68 % of consecutive samples are exact duplicates. Game engines serve vehicle dynamics and software testing [33], but without measured road profiles they provide neither a valid IRI ground truth nor transferable vibration.

VII. Discussion

What simulated drives over measured profiles provide. Ranking segments within a route turned out to be the easy part, which an untrained, speed-normalised vibration RMS does nearly as well as the best trained estimators. Simulation supplied the mapping from vibration to IRI up to a level that depended on two simulator choices, the sensor path, whose effect reached the estimator through the features carrying vibration above 10 Hz, and the vehicle population; no variant was built to reproduce the absolute gain of this car's vibration path, and the reference simulator's matched level was a fortunate combination of these choices.

A blueprint for road agencies. For a new car type the results suggest three steps. (i) Simulate the car over corridors whose profiles are already surveyed, with randomised vehicles and a sensor model; the ranking of segments will transfer. (ii) Collect about 5 km of paired drives over profiled road; one pass per segment sufficed here. (iii) Train jointly on simulated and paired drives, which beat real-only training on average for every simulator variant and every k we tested, though not on every route and draw; a level offset is slightly more accurate when the simulated level error is common to both road classes, which the paired drives reveal by class and a comparison of real and simulated spectra on the calibration roads anticipates (an association found after the fact).

How to evaluate. Because a within-route Spearman correlation is unaffected by calibration, it cannot separate a trained estimator from an untrained statistic; IRI estimators should be judged on held-out routes by absolute error and level per road class. Segment errors are spatially correlated: 5 km-block intervals of paired differences were 1.9–2.7 times as wide as segment-level ones, and route-level intervals with four routes include zero for most comparisons (Table S5), so the supported differences describe these four routes, not a population of routes, and benchmarks on road data should resample spatial blocks or routes, not segments. Smooth roads are driven fast, so estimators trained on real cars can use speed as a cue for low roughness, consistent with their over-correction on a fixed road; we recommend reporting such fixed-road elasticities from repeated passes, with the within-pass speed variability as a control, instead of speed sweeps in simulation alone. All errors are relative to one P79 survey (a second device: Table S11).

VIII. Limitations

External validity. LiRA-CD contains one car model with one type of telematics unit on four routes in one city region; road class and route are confounded, and with four routes no statement about route-to-route variation is inferential. Whether the simulated mapping transfers up to a level, and how much paired data the level needs, for other cars, sensors, speeds and road networks is unknown. Transfer to other vehicles and to phones in holders was examined only in simulation (Supplement S7). *Exploratory analyses.* Except for the comparison of the reference simulator with real training, all real-data results come from analyses designed after the test routes had been evaluated; they generate hypotheses rather than confirm them. *Combination methods.* We compared three ways of combining simulated and real data with one learner and fixed settings; network fine-tuning or domain adaptation might rank differently. *Simulator.* The simulator is a linear half-car without roll or body flexibility, driven by the mean of the two wheel-path profiles, with parameter ranges for sensitivity rather than a calibrated fleet; its vibration differs from the real car's at 2–5 Hz and above 10 Hz and in its speed scaling. *Other.* M7 is not a reproduction of [12] and the CNN was not tuned; the reference surveys preceded the car passes by two to fifteen months, segments are 100 m long, and we did not train on real data from several cities.

IX. Conclusion

Simulated drives over measured road profiles teach an estimator how a production car's vibration ranks road roughness, but not its absolute level, which depends on the simulated sensor path above 10 Hz and on the simulated vehicle population; an untrained vibration statistic ranks the segments nearly as well, so the informative test is the level. About 5 km of paired drives with one pass per segment fix the level: combined with simulation over all profiled roads, they beat real-only training on the same drives for every simulator variant, the reference simulator with a level offset beats 20 km of real drives alone, and joint training keeps an advantage for every variant up to 20 km, whereas a level offset loses it for the variants with the largest vibration deficit at 10–15 Hz. Game-engine road meshes cannot replace measured profiles. Whether these requirements hold for other cars and road networks needs new paired data, ideally with the simulator fixed in advance.

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